Jorge Mifsut Benet

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Research Interests

Probabilistic PDE forecasting, scientific machine learning, stochastic interpolants, generative models, uncertainty quantification, and complex dynamical systems, with applications to weather, physical, astrophysical, and cosmological simulation.

Education

Sorbonne Université (SU) — Paris, France
PhD, Deep-learning surrogate models for Modeling Spatio-Temporal Dynamics — *Nov. 2023 - Present*

Chalmers University of Technology (CTH) — Göteborg, Sweden
MSc, Complex Adaptive Systems (Engineering Physics) — *Aug. 2021 - Jun. 2023*

Universitat de Valencia (UV) — Valencia, Spain
BSc, Physics — *Sep. 2016 – Jul. 2021*

Université Paris Cité — Paris, France
Erasmus+ Exchange Program — *Sep. 2018 – Jun. 2019*

Research Experience

PhD Thesis — Nov. 2023 – Present
Institut des Systèmes Intelligents et Robotique, SU

- Develop probabilistic deep-learning surrogate models for spatio-temporal partial differential equations, with a focus on uncertainty quantification and stable long-horizon forecasting.
- Investigate stochastic interpolants, latent stochastic dynamics, neural operators, and generative models for complex dynamical system modeling.

Master's Thesis — Dec. 2022 – Jul. 2023
Department of Earth, Space and Environment, CTH

- Developed a convolutional neural network to infer the protostellar mass and viewing inclination angle of massive young stellar objects from synthetic observations at 19 μm and 37 μm.
- Applied the model to observations of *Cepheus A* from the SOFIA Massive Star Formation Survey, obtaining estimates consistent with independent results reported in the literature.

Research Internship — Jul. 2022 – Aug. 2022
I2SysBio, Institute for Integrative Systems Biology, CSIC–UV

- Applied statistical learning and large-scale genomic analysis to investigate similarities between bacteriophages and their bacterial hosts.
- Developed data-analysis and visualization pipelines that contributed to a peer-reviewed publication in *Viruses*.[1]

Publications

[1] Vicente Arnau, Wladimiro Díaz-Villanueva, **Jorge Mifsut Benet**, Paula Villasante, Beatriz Beamud, Paula Mompó, Rafael Sanjuan, Fernando González-Candelas, Pilar Domingo-Calap, and Mária Džunková. "Inference of the Life Cycle of Environmental Phages from Genomic Signature Distances to Their Hosts". In: *Viruses* 15.5 (2023). ISSN: 1999-4915 URL.

[2] Armand Kassaï Koupaï, **Jorge Mifsut Benet**, Yuan Yin, Jean-Noël Vittaut, and Patrick Gallinari. "Boosting Generalization in Parametric PDE Neural Solvers through Adaptive Conditioning". In: *The Thirty-eighth Annual Conference on Neural Information Processing Systems.* 2024 URL.

[3] **Jorge Mifsut Benet**, Armand Kassaï Koupaï, Ramon Daniel Regueiro-Espino, Nicolas Baskiotis, and Patrick Gallinari. "Adaptive SDE Interpolants for Calibrated Probabilistic PDE Forecasting". In: *AI&PDE: ICLR 2026 Workshop on AI and Partial Differential Equations.* 2026 URL.

Invited Talks

Uncertainty Quantification in Physics Transformers Saint-Étienne, France
ANR MELISSA kick-off *May 12, 2025*

Academic Service

Reviewer: AI&PDE Workshop at ICLR 2026

Technical Skills

Machine learning: PyTorch, JAX, neural operators, generative modeling, uncertainty quantification
Scientific computing: NumPy, SciPy, PDE simulation
Infrastructure: Slurm, multi-GPU training, Linux, Git, Weights Biases
Programming: Python, C/C++, MATLAB
Typesetting: LaTeX

Volunteering

Course Representative 2022 – 2023
Chalmers University of Technology
International Student Mentor 2019 – 2021
Universitat de València